

Feature Exposure

Measuring reliance

A model's overall correlation with target tells you how good predictions are but nothing on how it got there. Whether it's spread across many features $\rightarrow$ ROBUST or concentrated on a few $\rightarrow$ FRAGILE.

Evidence

Across 80+ models: inverse relationship when low in-sample max feature exposure (with Spearman rank corr.) reliably higher out-of-sample performance.

⚠ Low exposure, before seeing new data, predicted robustness to non-stationarity (regime change).